

Data Science

CSE558

Recap

Attribute types

- Nominal (Name, State)
- Ordinal (Grade, Class)
- Discrete (Months, Days)
- Continuous (π , 6.786)

Overview

- Attribute types
- Statistical measures
 - Measure of central tendency & Dispersion
 - Correlations
- Plots

Encodings

Most model deals with numbers!

Continuous

Ordinal: Assign integers

Discrete: One-hot encoding

Nominal: Embeddings

Questions

Mixed data!

Too many discrete attributes!

Large feature space

Statistics Everywhere

Highest (Maximum) marks, speed, ...

Lowest (Minimum) unemployment rate, mortality rate, ...

Average age, height, ...

Stock market fluctuation (Variance)

Chance (Likelihood) of rain tomorrow

If rain then more traffic (Correlation)

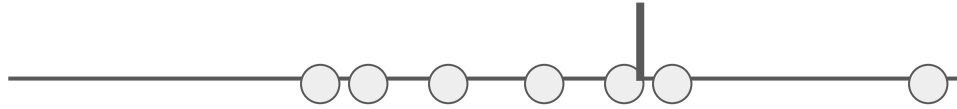
On average do indian spends more on lunch or dinner (t-test)

Overview

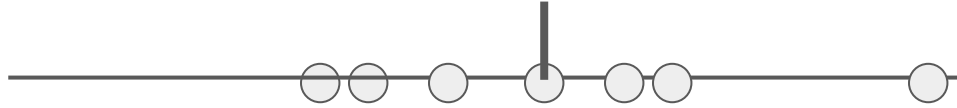
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Measure of Central Tendency

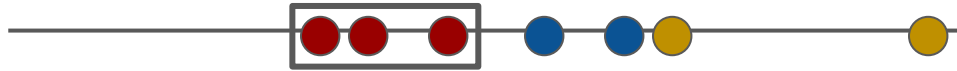
Mean:



Median:



Mode:



Mean - Average

Let $x_1, x_2, \dots, x_N$ be N values or observations for some numeric attribute. The **mean** of these values is

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + \dots + x_N}{N}$$

If every value x_i in the set is associated with a weight w_i for $i = 1, \dots, N$. The **weighted mean** is

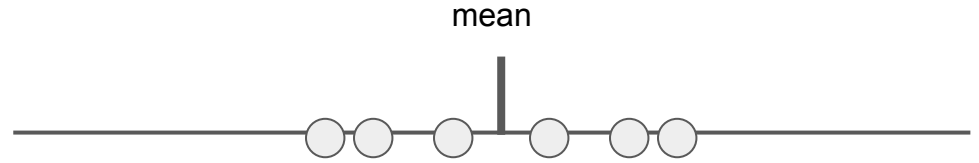
$$\bar{x} = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i} = \frac{w_1 x_1 + w_2 x_2 + \dots + w_N x_N}{w_1 + w_2 + \dots + w_N}$$

Example

Values: 100, 200, 300

Mean = $(100+200+300)/3 = 600/3 = 200$.

Used on continuous attribute.



Running time? $O(N)$

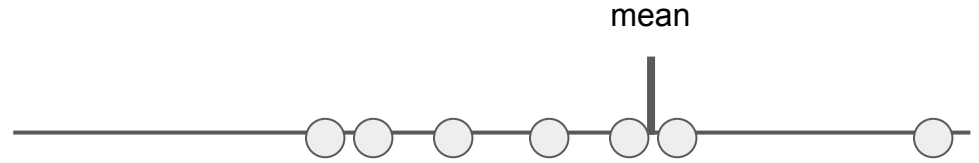
Any Problem?

Example

Values: 100, 200, 300, 10000

Mean = $(100+200+300+10000)/4 = 10600/4 = 2650$.

Sensitive to outliers.



Solutions?



Truncated Mean

Truncated Mean

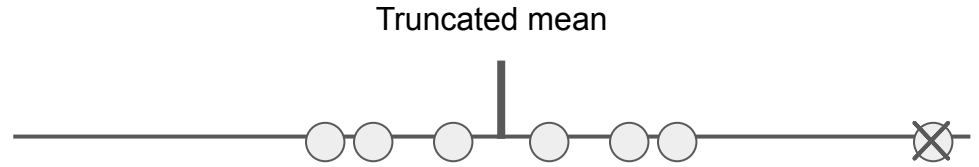
Values: 100, 200, 300, 10000, 1

Sorted Values: 1, 100, 200, 300, 10000

Truncated Mean = $(100+200+300)/3 = 600/3 = 200$.

Running time?

$O(N \cdot \log(N))$



Median - Mid term/value

Values: 100, 200, 300, 10000, 1

Sorted Values: 1, 100, 200, 300, 10000

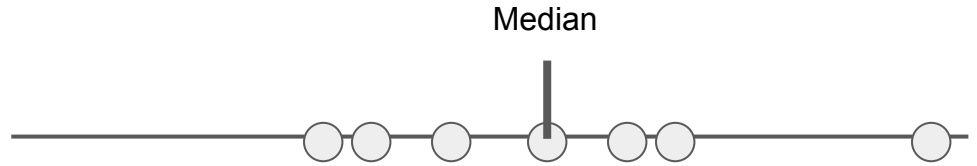
Median: 200.

Used on ordinal attribute & continuous attribute.

Running time?

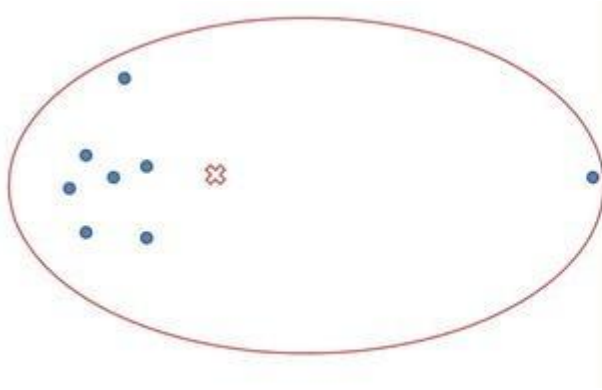
$O(N \cdot \log(N))$

Any faster?

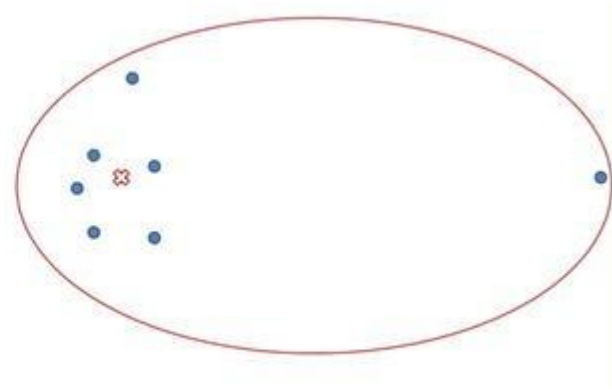


Median - Mid term/value

Medoids: Median for higher dimension data.



Mean



Medoid

Mode - Most Frequent

It is the most common value.



Used on nominal attribute, ordinal attribute and discrete attribute,

Measures of dispersion

Let $x_1, x_2, \dots, x_N$ be N values or observations for some numeric attribute. The **variance** of these values is

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$$

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N}$$

The **standard deviation**, $\sigma = \sqrt{\sigma^2}$

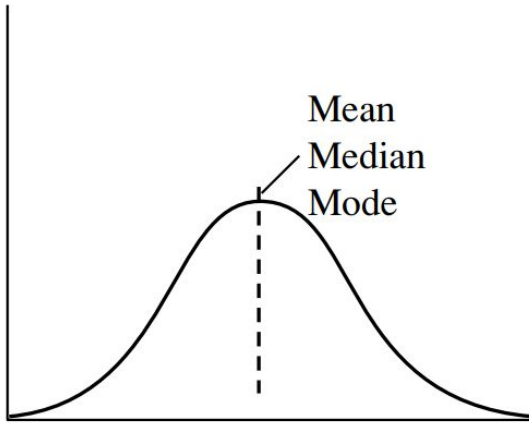
Mean absolute deviation $\frac{1}{N} \sum_{i=1}^N |x_i - \bar{x}|$

Overview

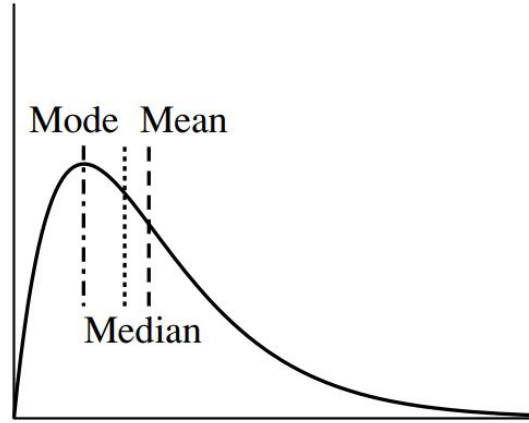
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Skewness

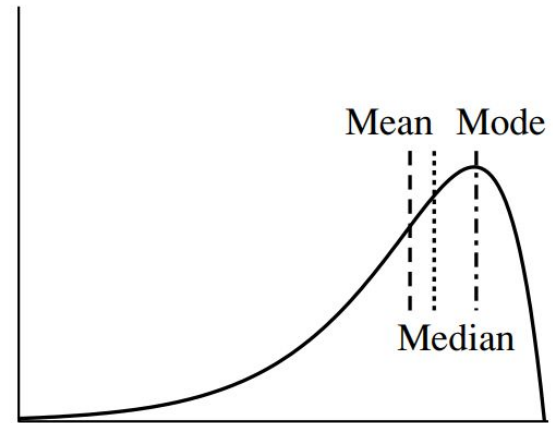
Measures nature of symmetry



Symmetric



Positively Skewed



Negatively Skewed

Karl-Pearson Coefficient

Mode available:

$$kp_1 = \frac{\bar{x} - \text{Mode}}{\sigma}$$

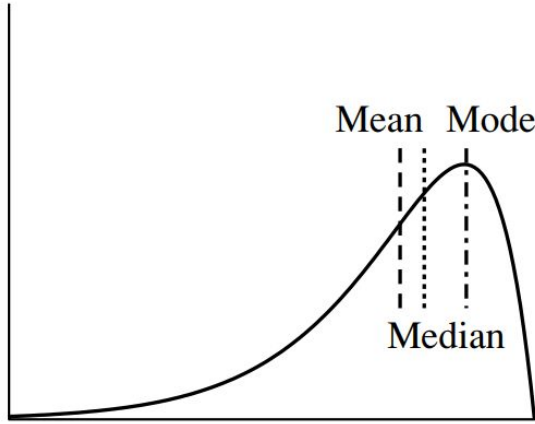
within (-1, +1)

Mode unavailable:

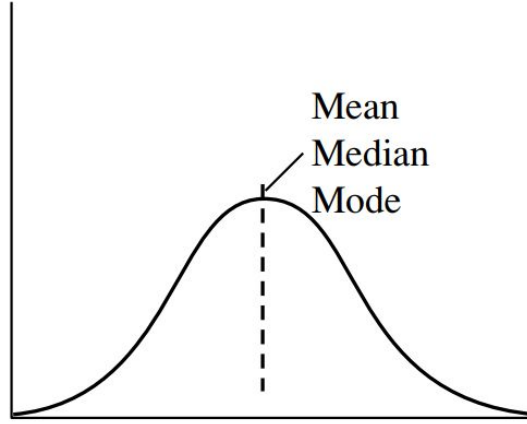
$$kp_2 = \frac{3(\bar{x} - \text{Median})}{\sigma}$$

within (-3, +3)

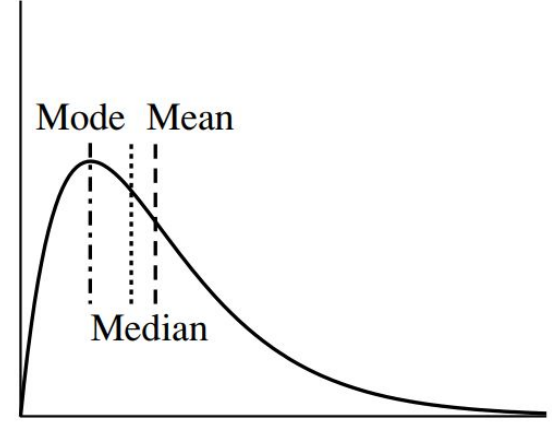
Karl-Pearson Coefficient



Coefficient < 0



Coefficient $= 0$



Coefficient > 0

Correlation

Does one variable depends on the other?

- Remove outliers if any.
- Check if distribution is normal or close to normal.

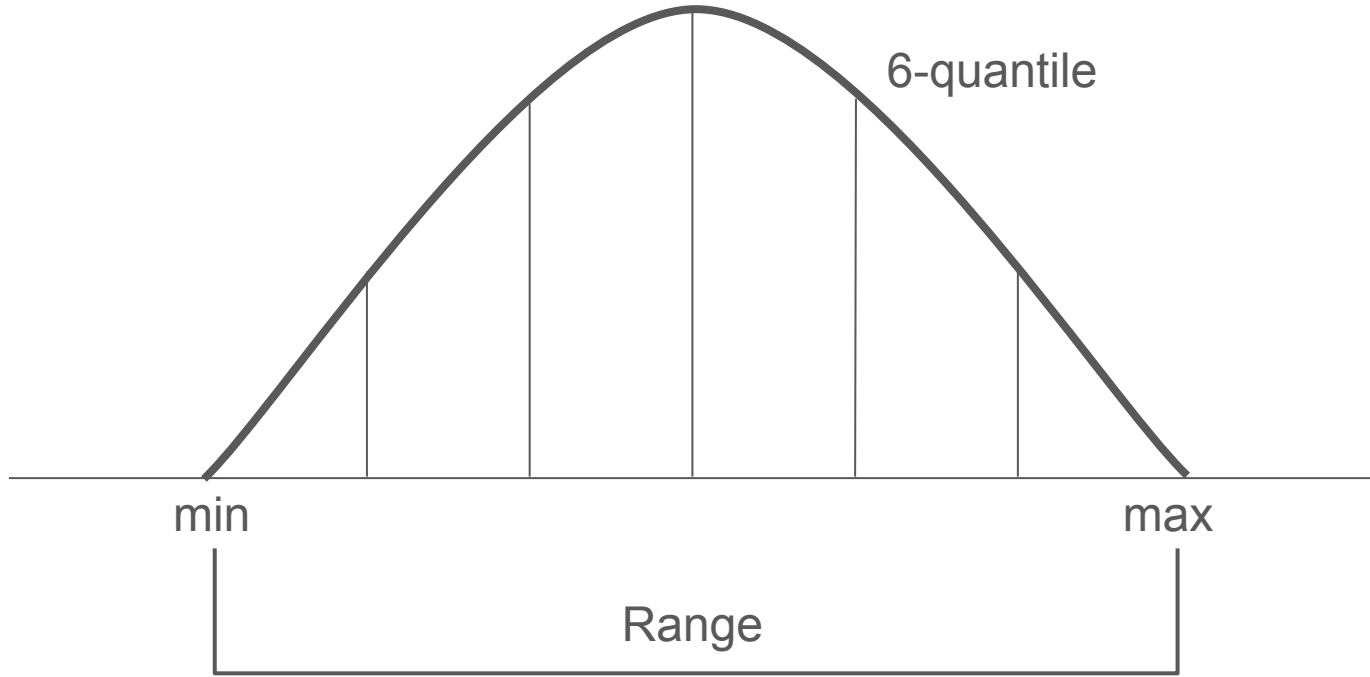
$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$$

Does this imply causation?

Overview

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 - Measure of central tendency & Dispersion
 - Skewness
 - Correlations
- Plots

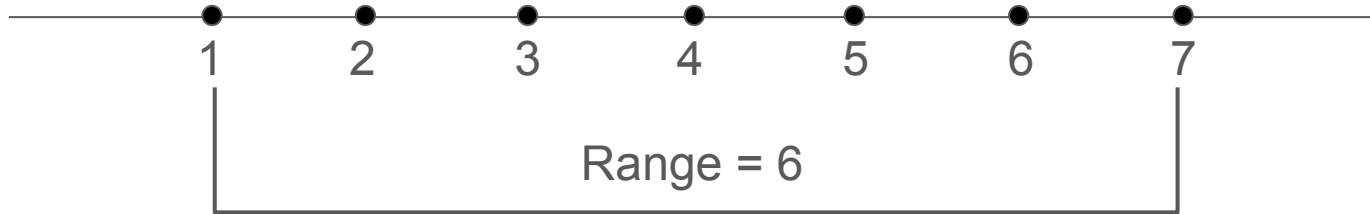
Range & Quantile



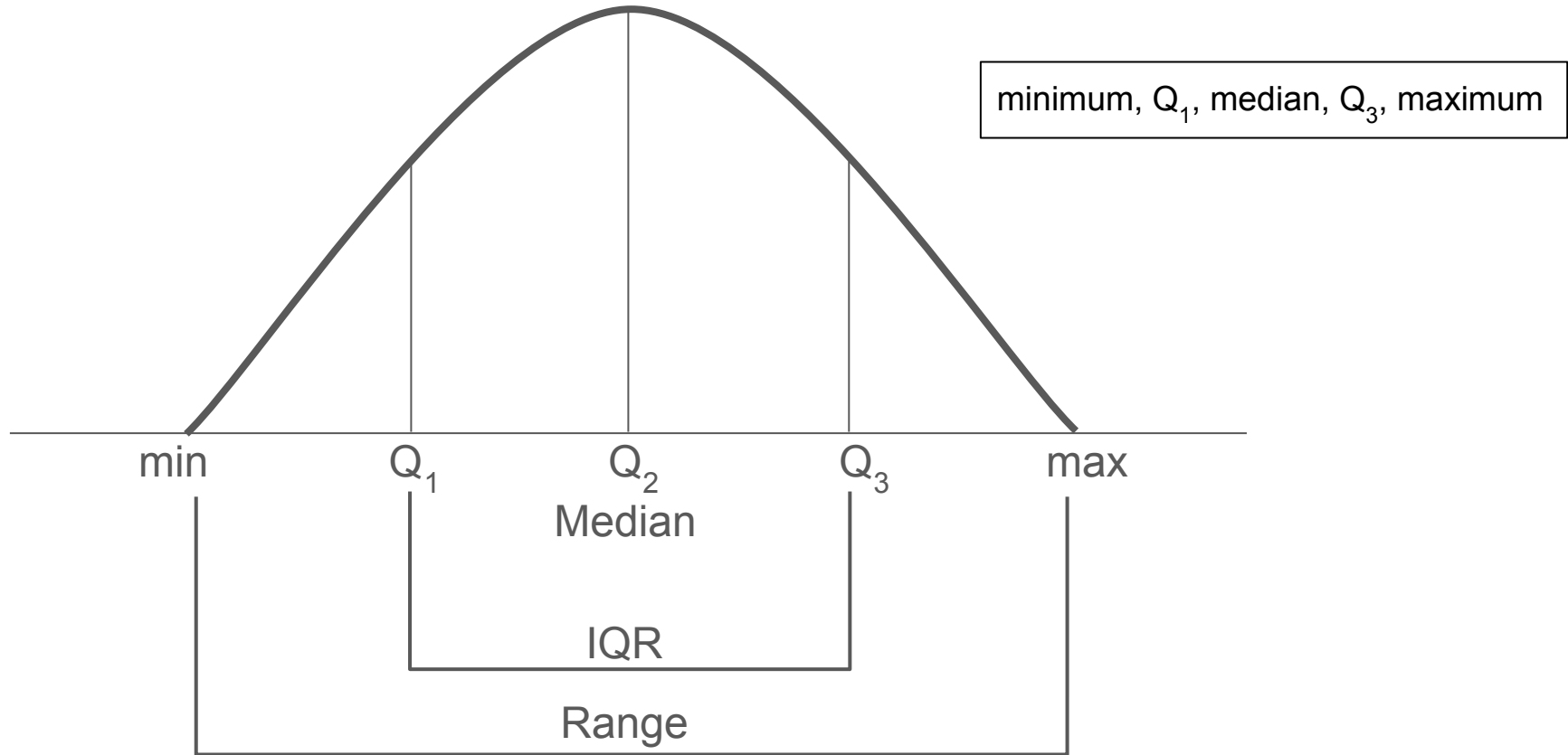
$$\text{Range} = \text{max}() - \text{min}()$$

Example

Values: 1, 3, 4, 2, 7, 6, 5



Summary from Quartile



Example

25, 80, 35, 30, 60, 20, 65, 70, 45, 85, 90, 75

Sort: 20, 25, 30, 35, 45, 60, 65, 70, 75, 80, 85, 90

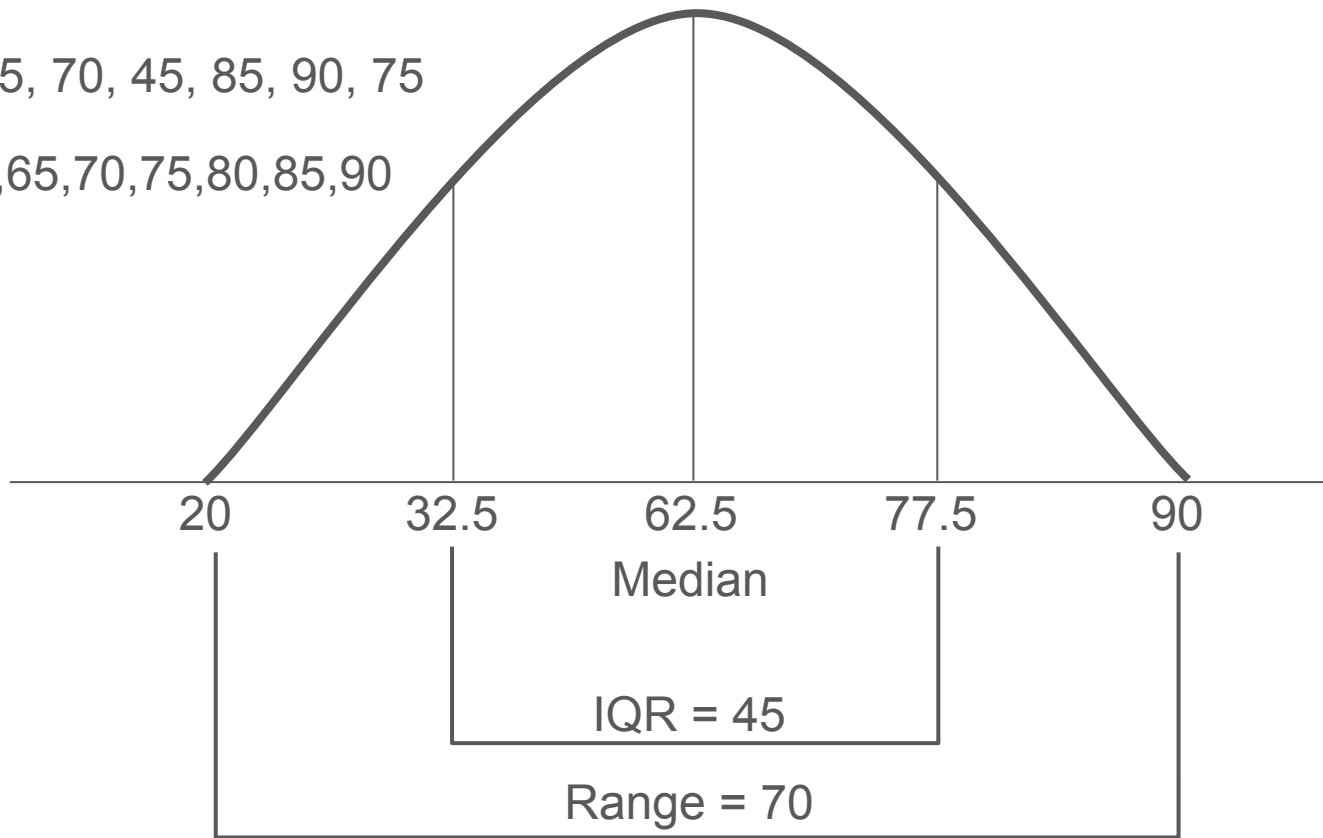
Minimum = 20

$Q_1 = 32.5$

$Q_2 = 62.5$

$Q_3 = 77.5$

Maximum = 90

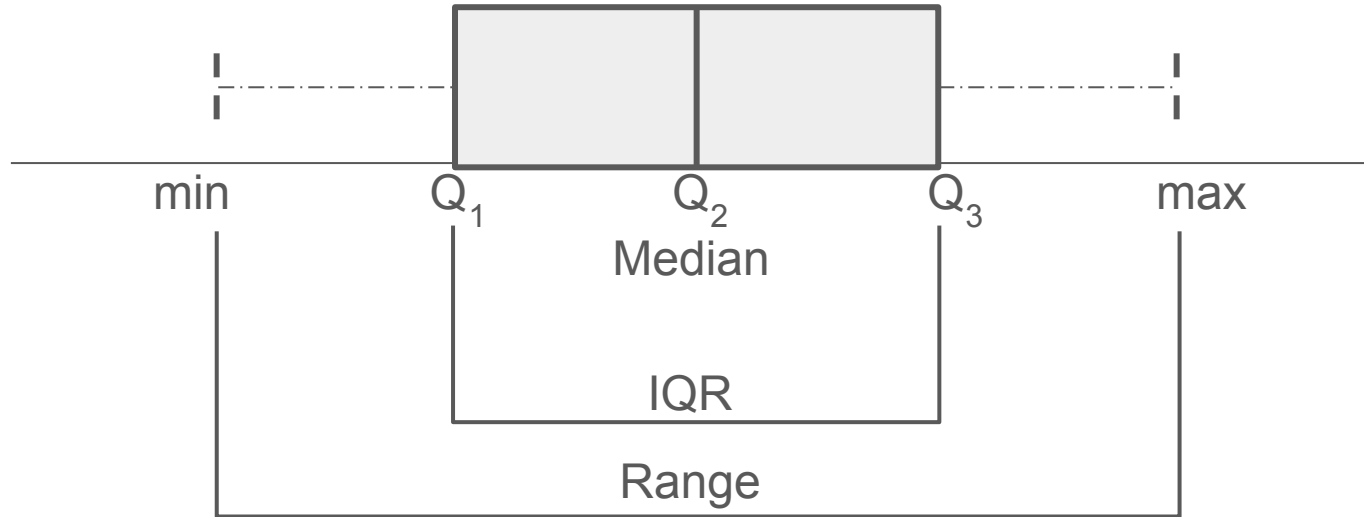


Boxplot

Simple graphics to realize distribution

Indicates skewness in data

Can be used to identify outliers



Moments

Moments about origin, i.e., 0

First moment $\frac{1}{N} \sum_{i=1}^N (x_i - 0)$

Second moment $\frac{1}{N} \sum_{i=1}^N (x_i - 0)^2$

r^{th} moment $\frac{1}{N} \sum_{i=1}^N (x_i - 0)^r$

Moments about mean, i.e., $\bar{x}$

First moment $\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})$

Second moment $\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$

r^{th} moment $\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^r$

Properties

First moment about 0 is mean $\frac{1}{N} \sum_{i=1}^N (x_i - 0)$

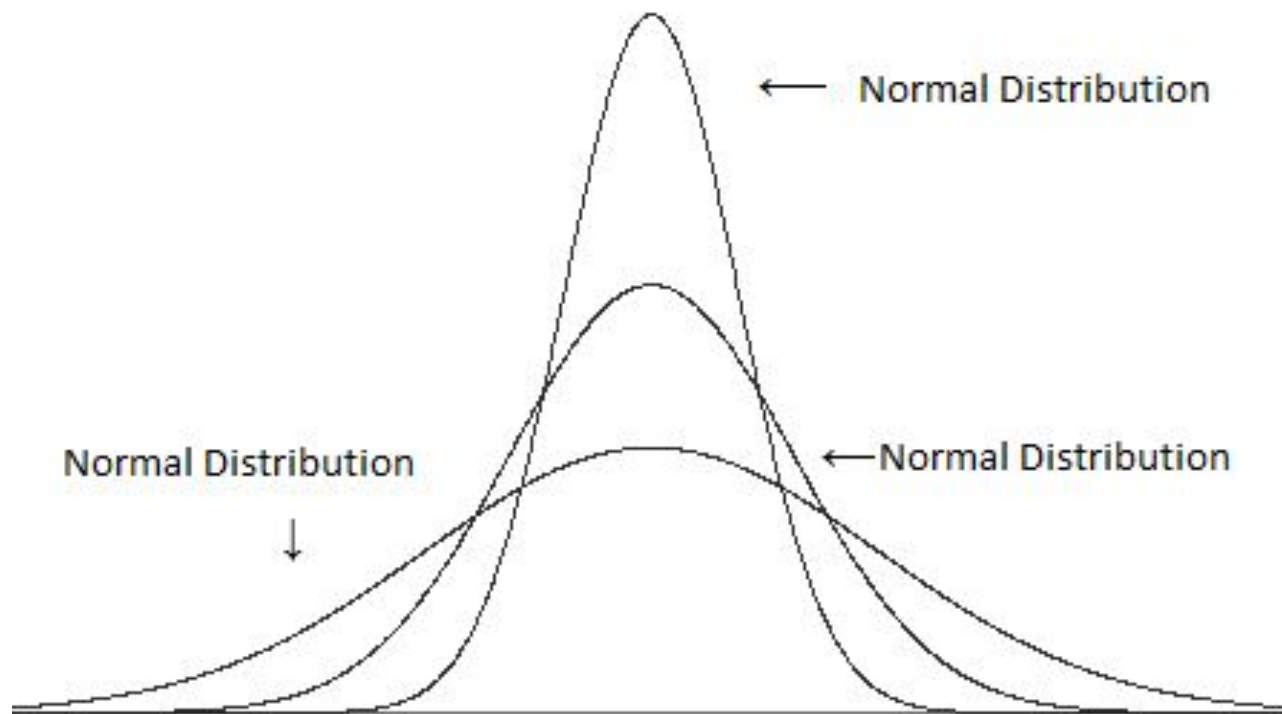
Second moment about $\bar{x}$ is variance $\frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$

Third moment about $\bar{x}$ is used to measures skewness. $\mu_3 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^3$

Pearsons' moment coefficient of skewness $\gamma_1 = \frac{\mu_3}{\sigma^3}$

Kurtosis

Measures the flatness



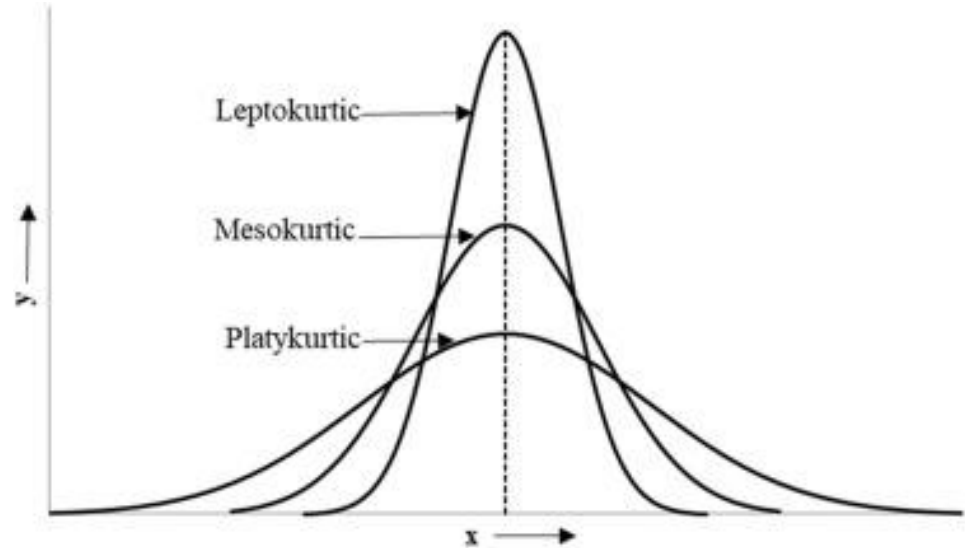
Kurtosis Coefficient

$$\gamma_2 = \frac{\mu_4}{\sigma^4}$$

Leptokurtic > 3

Mesokurtic $= 3$

Platykurtic < 3



Quantile Plot

Visualizes univariate data distribution.

1. Sort data
2. Compute f_i score.

$$f_i = \frac{i - 0.5}{N}$$

Example

<i>Unit price (\$)</i>	<i>Count of items sold</i>
40	275
43	300
47	250
—	—
74	360
75	515
78	540
—	—
115	320
117	270
120	350

